Machine Learning for Mechanical Engineers at CoEP Session 8: Linear Regression

Session 8: LinRegr

Linear Regression

Want to know future? Predictions

- How does sales volume change with changes in price?
- How is this affected by changes in the weather?
- How does the amount of a drug absorbed vary with dosage and with body weight of patient?
- Does it depend on blood pressure?

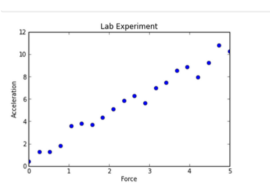
Answering questions like these, requires us to create a model.

The model

- A model is a formula where one variable (response) varies depending on one or more independent variables (covariates).
- The formula (model) can be simple or complex.
- Simplest is a Linear Model where the dependent varies linearly with the independent variable(s).
- Creating a Linear Model involves a technique known as Linear Regression.

Remember? - High School Physics Lab

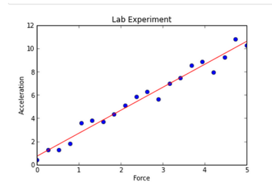
- Finding how acceleration changes when Force is applied
- For some input X (say force), got output Y (say acceleration).
- Plotted the pairs of observations x, y.



What next?

Fitting line

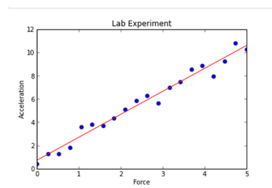
- Then you had to fit a straight line through points
- A visual "best fit", some points above, some below.
- Found 'm', the slope (how?), and 'b', the intercept (how?).



Whats the model here?

Fitting line

- Equation of line, the formula, itself is the model.
- You were doing informal Linear Regression.



Linear Regression

- What: for predicting a quantitative variable
- Age: "it's been around for a long time". The term regression was devised by Francis Galton in his article Regression Towards Mediocrity in Hereditary Stature in 1886.
- Complexity: easy to understand and use
- Popularity: still widely used

Possible Techniques

- Statisticians know Regression as OLS (Ordinary Least Squares), a method to fit a line. It minimizes the sum of the squared residuals.
- Theoretical/Analytical Solution $W = (X^TX)^{-1}X^Ty$. Not practical for large matrices.
- Also, X^TX cannot be inverted if any two columns (features) are identical, or one is a combination of others.
- Machine Learning Engineers know Regression as prediction of continuous variables (floats). It uses Gradient Descent for the minimization part.

Intuition
Both routes look for the same line. The analytical route jumps straight to the answer with one formula, like solving an equation on paper. Gradient Descent walks towards it step by step, which is slower per line but keeps working when the dataset is far too large to invert a matrix.

Linear Regression - Analytical Approach (OLS)

Example: Tip Amount

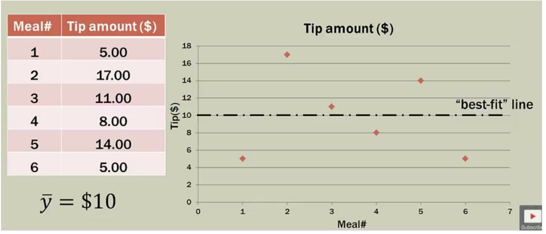
- In a restaurant, tip you pay to the waiter,
- Typically depends on the bill amount.
- Smaller the bill, lesser the tip, and vice versa.

Can we predict the tip amount just by looking at the bill amount?
Inputs?
Outputs?

Example: Tip Amount

- You collect, random (meaning, any) 6 tips.
- YOU HAVE NOT COLLECTED THE BILL AMOUNT.
- Just one variable (Meal id is just the descriptor its not a independent variable)
- Any guesses?

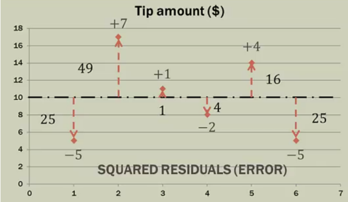
Most Basic Situation



- With just one variable, mean is the best we can do.
- Note: not a single point falls on it.
- Its just an estimate.
- Goodness of fit?

(Reference: Simple Linear Regression: Step-By-Step - Dan Wellisch)

Example: Tip Amount

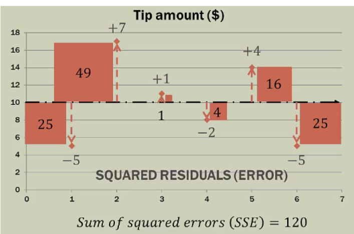


Meal#	Residual	Residual²
1	−5	25
2	+7	49
3	+1	1
4	−2	4
5	+4	16
6	−5	25

Sum of squared errors (SSE) = 120

- Std Variation says about goodness of fit.
- Diff of each point with mean are called ‘residuals’ or ‘errors’
- Idea is to minimize SSE (Sum of Squared Errors) or RSS (Residual Sum of Squares)

Important Concept



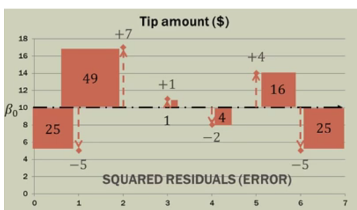
When conducting simple linear regression with TWO variables, we will determine how good that line "fits" the data by comparing it to THIS TYPE; where we pretend the second variable does not even exist.

If a two-variable regression model looks like this example, what does the other variable do to help explain the dependent variable?

NOTHING.

- With just the target variable (tips) we got SSE as 120.
- If we add one more variable to the problem, ie bill amount, what do you expect of SSE?
- It should reduce, or else, whats the use of adding this variable.
- So, we will compare our Linear regression fit line with this (one variable) line.

Important Concept



When conducting simple linear regression with TWO variables, we will determine how good the regression line "fits" the data by comparing it to THIS TYPE; where we pretend the second variable does not even exist; the slope, $\beta_1 = 0$.

In this situation, the value of $\hat{y}$ is 10 for every value of x.

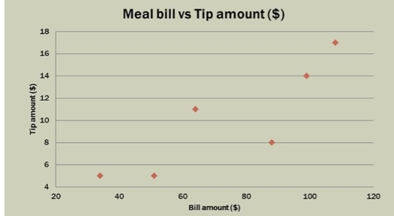
$\hat{y} = b_0 + b_1 x$ $\quad b_0 = 10$

$\hat{y} = b_0 + (0)x$

$\hat{y} = b_0$ $\quad \hat{y} = 10$

- We find slope and intercept as b_1 and b_0 respectively.
- A line with $b_1 = 0$ is the horizontal line.
- Its SSE becomes the benchmark. Any selected pair of b_1 and b_0 should give SSE less than the benchmark SSE.

Linear Regression with Two Variables

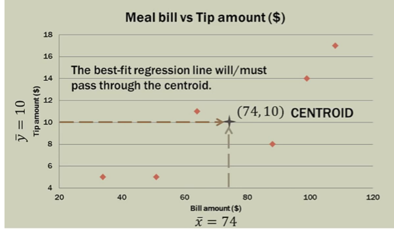


Bill ($)	Tip ($)
34.00	5.00
108.00	17.00
64.00	11.00
88.00	8.00
99.00	14.00
51.00	5.00

- You can fit a line, however you want.
- Even horizontal line can be put
- But then, which one is the best?
- Need to be at least better than horizontal line!!

Methodology

Hypothesis: A line passing through the centroid of the data points will be good. Lets check.



Bill ($)	Tip ($)
34.00	5.00
108.00	17.00
64.00	11.00
88.00	8.00
99.00	14.00
51.00	5.00
$\bar{x} = 74$	$\bar{y} = 10$

- Calculate mean of both x and y.
- Line needs to pass through this point

Calculations

Intercept $\qquad \hat{y}_i = b_0 + b_1 x_i \qquad$ Slope

$$b_0 = \bar{y} - b_1\bar{x} \qquad b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$\bar{x}$ = mean of the independent variable $\quad x_i$ = value of independent variable

$\bar{y}$ = mean of the dependent variable $\quad y_i$ = value of dependent variable

- Intercept and Slope are calculated as above.
- A simple tabular calculations can be done.

Calculations

Meal	Total bill ($)	Tip amount ($)	Bill deviation	Tip Deviations	Deviation Products	Bill Deviations Squared
	x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	34	5	-40	-5	200	1600
2	108	17	34	7	238	1156
3	64	11	-10	1	-10	100
4	88	8	14	-2	-28	196
5	99	14	25	4	100	625
6	51	5	-23	-5	115	529
	$\bar{x} = 74$	$\bar{y} = 10$			$\sum = 615$	$\sum = 4206$

- Slope comes to $= 615/4206 = 0.1462$
- Intercept comes to $= 10 - 0.1462(74) = -0.8188$
- So, eq is $y = 0.1462x - 0.8188$

Interpretation

$$\hat{y}_i = 0.1462x - 0.8188$$

For every $1 the bill amount ($x$) increases, we would expect the tip amount to increase by $0.1462 or about 15-cents.

If the bill amount (x) is zero, then the expected/predicted tip amount is $-0.8188 or negative 82-cents! Does this make sense? NO. The intercept may or may not make sense in the "real world."

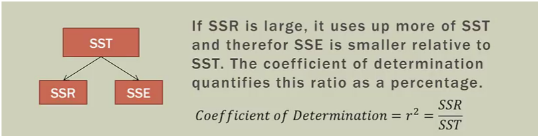
- Slope gives proportionality with which tip is calculated.
- Intercept may or may not make sense in real life.
- Is this manual calculation giving good line?

Evaluation: SSE of Our Model

Meal	Total bill ($)	Observed tip amount ($)	$\hat{y}_i$ (predicted tip amount)	Error $(y - \hat{y}_i)$	Squared Error $(y - \hat{y}_i)^2$
	x	y			
1	34	5	4.1505	0.8495	0.7217
2	108	17	14.9693	2.0307	4.1237
3	64	11	8.5365	2.4635	6.0688
4	88	8	12.0453	−4.0453	16.3645
5	99	14	13.6535	0.3465	0.1201
6	51	5	6.6359	−1.6359	2.6762
	$\bar{x} = 74$	$\bar{y} = 10$		SSE=	$\sum = 30.075$

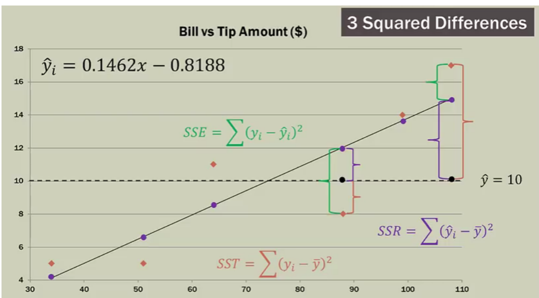
- Sum of Squares of Error , SSE is 30 for our manual model
- Its far less than the horizontal line benchmark of 120.
- So, by adding Bill amount as independent variable, reduced error by about 90.

Evaluation: SST, SSE and SSR



- SST (sum of squares total) is the benchmark error. Its constant ie 120.
- SSE (sum of squares error) is the level to which we could get error to it by applying regression ie 30.
- The difference, the achievement by regression is called SSR (sum of squares by regression) ie 90.

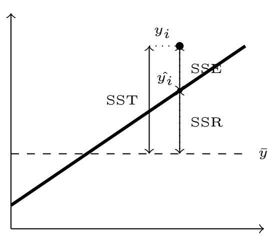
Evaluation: Three Distances



- For each x, there is predicted y and actual y
- 3 relationships: observed, predicted and total
- Can decide quality based on the ratios among them.

R^2 Statistics

- $SST = SSR + SSE$
- SST: total variability in the response, before any regression. Measured y_i to $\bar{y}$.
- It splits into two parts, one explained by the model, one not.
- SSR: the variability the regression explains. Measured $\hat{y}_i$ to $\bar{y}$.
- SSE: the variability left unexplained. Measured y_i to $\hat{y}_i$.



One data point, three vertical gaps. The total gap splits into the part the line explains and the part it does not.

Intuition

SST is how wrong you were before drawing any line, using only the mean. SSE is how wrong you still are after drawing it. The difference, SSR, is what the line actually bought you.

R^2: Putting a Number on the Fit

- R^2 is the fraction of the total variability that the regression explains.

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- For our tip example: $SST = 120$, $SSE = 30$, so $SSR = 90$.
- $R^2 = 90/120 = 0.75$. The bill amount explains 75% of the variation in tips.
- $R^2 = 1$ is a perfect fit, $R^2 = 0$ means the line is no better than the mean.

Intuition

R^2 is a score out of 1 for the question "was adding this variable worth it?". The horizontal mean line always scores 0, so anything above 0 is progress. It is a ratio, so it stays comparable across problems measured in completely different units.

Quick Check: Least Squares

Think About It

We fitted the tip line by forcing it through the centroid of the data, the point $(\bar{x}, \bar{y})$. Suppose a waiter adds one more meal to the records: a huge bill with a tip of exactly zero. What happens to the fitted line, and what does that tell you about least squares?

Quick Check: Least Squares

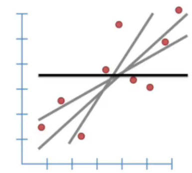
Answer

The line swings noticeably towards that one point. Because the method squares each error, a point far from the line contributes enormously more than a point near it: an error of 10 counts 100 times as much as an error of 1. So a single outlier can drag the whole fit, and SSE jumps while R^2 falls. This is why we plot the data before trusting a fit, and why outliers get checked rather than silently kept.

Linear Regression - Machine Learning Approach

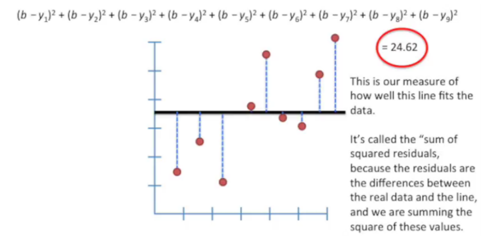
Linear Regression

- We have multiple observations of inputs and an output
- Object: to find a model, ie a formulation, adhering to more or less all data points
- ie. to attempt to find the "best" line
- But, which line is the "best"?
- The horizontal line is the worst, the benchmark. We need to beat that!!



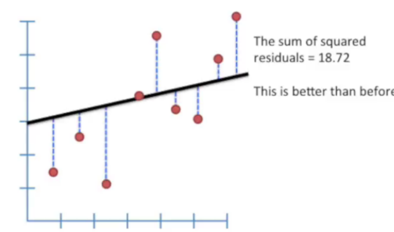
(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Benchmark Line



(Ref: StatQuest: Linear Models - Josh Starmer)

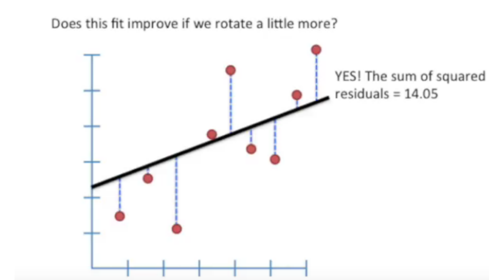
Sum of Squared Errors for Rotated Line



Its better but is this best?
(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Rotated Line

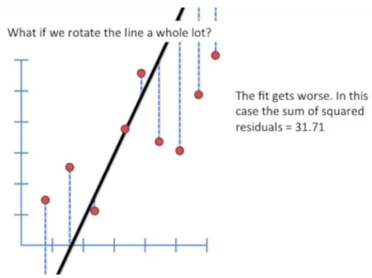
Rotated a bit more. Better?



Its better but is this best?
(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Rotated Line

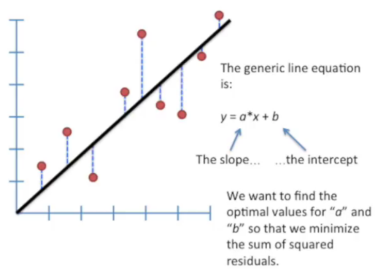
Why not rotate a whole lot!!



No, we lost the BEST point somewhere in between. Whats that "optimized" point?
(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization: Formulating the Problem

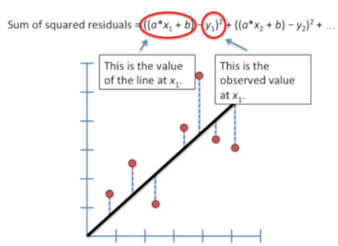
To find the sweet spot ie the optimized point, lets put some formulation based on generic line equation.



(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization: The Least Squares Objective

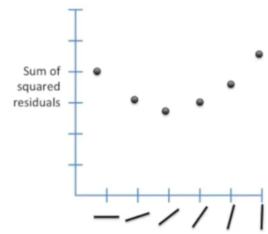
Mathematically . . .



Since we want the line that will give us the smallest sum of the squares of errors (SSE: Sum of Squared Errors), this method for finding the best values for "a" and "b" is called as "Least Squares".
(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization: The SSE Curve

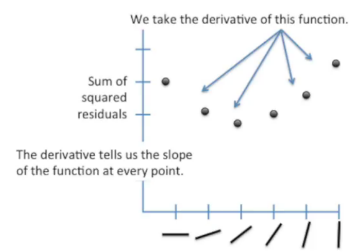
- If we plotted SSE for each rotated line the plot looks like.
- We can see that SSE was high for the horizontal line, then came down as we started rotating the line,
- But started rising again, if we passed the sweet spot.



(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization: Sliding Down the Curve

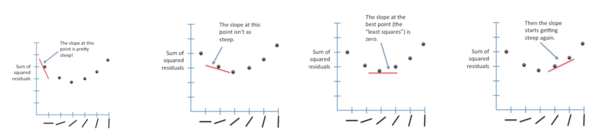
- We need to get to the bottom point.
- We can start at any (random) point, and SLIDE down to bottom.
- SLIDING means going along slope.
- Slope is given by Derivative of the (SSE) function/curve, at the point



(Ref: StatQuest: Linear Models - Josh Starmer)

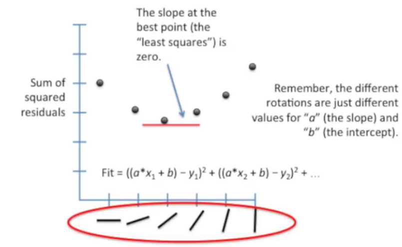
Optimization: Finding the Zero Slope

- Slope starts decreasing from end, faster, then slower,
- Reaches 0, horizontal, and the starts increasing again.
- Need to find the "0" slope point.



These different slopes mean different "a" parameters (Ref: StatQuest: Linear Models - Josh Starmer)

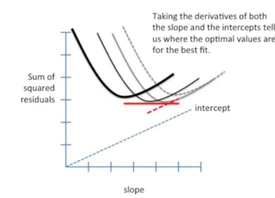
Optimization: Gradient Descent Steps



(Ref: StatQuest: Linear Models - Josh Starmer)

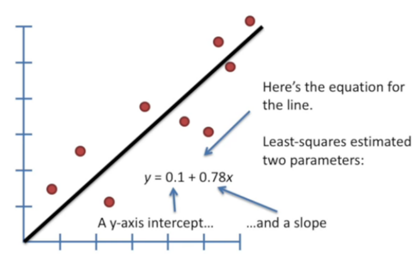
Optimization: Two Parameters, a Surface

- For "a" and "b" both to be optimized, we will need 3D picture, as the Loss function would then take Z axis
- It will be a Surface.
- Derivative will be Partial, but the objective is the same, Bottom-most point. ie Minimum Loss function value



The '"a" and "b" coordinates corresponding to the bottom-most point give most minimum Loss. So it represents the "best" line.
(Ref: StatQuest: Linear Models - Josh Starmer)

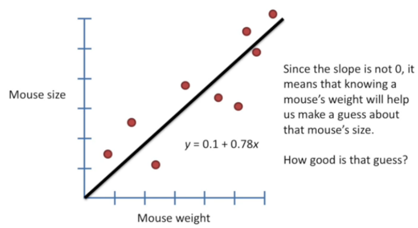
Results



(Ref: StatQuest: Linear Models - Josh Starmer)

Example: Mouse Size

- We have data points having mouse size and mouse weight
- We wish to find mouse size given mouse weight.



we use R^2 to find "good-ness" of fit (of line)
(Ref: StatQuest: Linear Models - Josh Starmer)

Linear Regression for Complex Problems

- In real life, problems are far more complex than just one x and one y.
- There could be many x's ie many inputs which decide the output
- General formulation : Predicting quantitative response y based on predictor variables x. Assumes linear relationship between x and y

$$y_i = \beta_0 + x_{1,i}\beta_1 + x_{2,i}\beta_2 + \cdots + x_{p,i}\beta_p + \epsilon_i$$

- Objective is to find the parameters β_j that minimize the squared residuals
- There are no slope formulas for these βs. So, we will solve the machine learning way.

Quick Check: Gradient Descent

Think About It

The analytical formula $W = (X^TX)^{-1}X^Ty$ gives the exact best line in one shot. Gradient Descent only creeps towards it, one step at a time, and may never land exactly on it. So why is Gradient Descent the method that actually runs in practice?

Quick Check: Gradient Descent

Answer

Because the formula requires inverting X^TX, and that becomes impossibly slow as the number of features grows, besides failing outright when two features are linearly dependent. Gradient Descent never inverts anything: it only needs the slope at the current point, so it scales to huge datasets. It also generalises, since the same walk-downhill idea trains models such as neural networks, where no closed-form formula exists at all. Being approximate is a price worth paying.

Linear Regression - Advertising Budget Example

Example: Advertising Dataset

- Toy Dataset: Advertising, Advertising.csv
- Sales totals for a product in 200 different markets. Advertising budget in each market, broken down into TV, radio, newspaper

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9

(Data: https://www.statlearning.com)

Problem

- Goal: What marketing plan for next year will result in high product sales?
- Questions (Hypothesis):
  - Is there a relationship between advertising budgets and sales?
  - How strong is the relationship between advertising budgets and sales?
  - Which media contribute to the sales the most?
  - Is the relationship linear?

EDA

Correlation Matrix

	TV	Radio	Newspaper	Sales
TV	1.000000	0.054809	0.056648	0.782224
Radio	0.054809	1.000000	0.354104	0.576223
Newspaper	0.056648	0.354104	1.000000	0.228299
Sales	0.782224	0.576223	0.228299	1.000000

- Correlation between radio and newspaper is 0.35
- Barely any correlation (or "not correlated") for TV/radio and TV/newspaper

EDA

	TV	Radio	Newspaper	Sales
TV	1.000000	0.054809	0.056648	0.782224
Radio	0.054809	1.000000	0.354104	0.576223
Newspaper	0.056648	0.354104	1.000000	0.228299
Sales	0.782224	0.576223	0.228299	1.000000

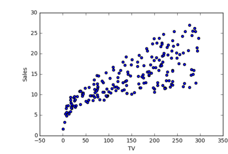
- Reveals tendency to spend more on Newspaper advertising in markets where more is spent on Radio advertising.
- Sales higher in markets where more is spent on Radio, but more also tends to be spent on Newspaper.
- In Simple LM: Newspaper "gets credit" for effect of Radio on Sales.

Advertising Dataset

- What are some ways we can regress sales onto advertising using Simple Linear Regression?
- One model:

$$sales \approx \beta_0 + \beta_1 \times TV$$
$$y \approx \beta_0 + \beta_1 \times x$$

- Scatter plot visualization for TV and Sales.

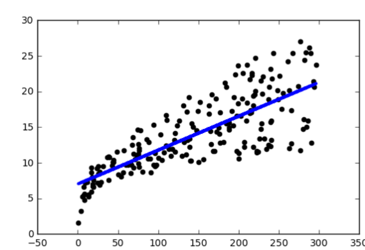


Advertising Dataset

- Simple Linear Model in Python (using pandas and scikit): Predictor: x, Response: y
- $Sales = 7.03259 + 0.04754 \times TV$

Advertising Dataset

Scatter plot visualization for TV and Sales with Linear Model.



If we have 3 such models, can we "somehow" compose them (say, averaging) to get the combined model?

Multiple Linear Regression

- In practice, often have more than one predictor
  - $sales \approx \beta_{0,1} + \beta_{1,1} \times TV$
  - $sales \approx \beta_{0,2} + \beta_{1,2} \times Newspaper$
  - $sales \approx \beta_{0,3} + \beta_{1,3} \times radio$
- Run three separate simple linear regressions
- For p predictor variables: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p + \epsilon$
- Since error ϵ has mean zero, we usually omit it.
- A one-unit change in any predictor variable x_i will change the expected mean response by β_j units. $sales = \beta_0 + \beta_1 \times TV + \beta_2 \times radio + \beta_3 \times newspaper$

Estimating the Parameters $\beta_0, \beta_1, \ldots$

- Advertising Dataset $sales = \beta_0 + \beta_1 \times TV + \beta_2 \times radio + \beta_3 \times newspaper$
- After solving $sales = 2.93 + 0.0457 \times TV + 0.188 \times radio - 0.0010 \times newspaper$

Coefficients

- Many Simple Linear Regressions, one per medium (slope, then intercept):
  - TV model: slope `0.04753664`, intercept `7.03259355`
  - Radio model: slope `0.20249578`, intercept `9.31163810`
  - Newspaper model: slope `0.05469310`, intercept `12.35140707`
- Single Multiple Linear Regression
  - Slopes: TV `0.04576465`, radio `0.18853002`, newspaper `-0.00103749`
  - Intercept: `2.93888937`
- Note the newspaper slope: clearly positive on its own, but essentially zero once TV and radio are in the model.

Intuition
On its own, newspaper spending looks like it drives sales. But markets that spend on newspaper also spend on radio, so newspaper was taking credit for radio's effect. Holding the other two fixed, newspaper's contribution collapses to nearly nothing. This is why the three separate regressions and the single combined one disagree.

Answering questions using Regression Model

Goal: What marketing plan for next year will result in high product sales?

- Question: Is there a relationship between advertising budget and sales?
- Answer: Yes, hypothesis testing shows that we can reject the null hypothesis that $B_{TV} = B_{RADIO} = B_{NEWSPAPER} = 0$

Quick Check: Multiple Regression

Think About It
Fitted on its own, newspaper advertising has a clearly positive slope. Fitted alongside TV and radio, its slope drops to roughly zero. Nothing about the data changed between the two fits. So should the marketing team cut the newspaper budget, and what would you check first?

Quick Check: Multiple Regression

Answer
Probably yes, but the reason matters more than the answer. The correlation matrix showed radio and newspaper spending move together at 0.35. Markets spending on newspaper were also spending on radio, so in the solo fit newspaper collected credit for radio's effect. Only the multiple regression, which holds the others fixed, separates them. What to check first is that correlation: a coefficient that changes sharply when you add other predictors is a signal that the predictors overlap, not that the data is faulty. Correlation between predictors is not proof that any one of them causes sales.

In conclusion

- Pros of Linear Regression Model:
  - Provides nice interpret-able results
  - Works well on many real-world problems
- Cons of Linear Regression Model:
  - Assumes linear relationship between response and predictors: Change in the response Y due to a one-unit change in Xi is constant
  - Assumes additive relationship. Effect of changes in a predictor Xi on response Y is independent of the values of the other predictors